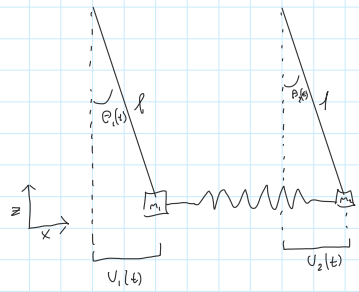
Force derivation

Wednesday, February 25, 2026 12:54 PM



mass 1

$$\boxed{F_g + F_t = F_{Pendulum}} \qquad \boxed{\vec{F}_s = F_{spring}}$$

mass 2

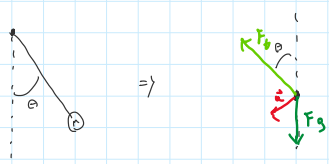
Since F_{net} is the Linear combination of all the Forces Present we can solve for the acceleration of these forces seperately and then combine them:

$$F_{net} = F_1 + F_2 + \dots F_n$$

$$\Rightarrow \cancel{m} a_{net} = \cancel{m} a_1 + \cancel{m} a_2 + \dots + \cancel{m} a_n$$

$$\Rightarrow a_{net} = a_1 + a_2 + \dots + a_n$$

Pendulum:



$\vec{F}_g$ is easy, if we assume any change in height will cause negligable change in $\vec{F}_g$ then we can say $\|\vec{F}\|$ is constant

lets assume we want a 1m Pendulum
the max height would be 2m (if it went all the way round)

$$\Delta F = G m_e m \left(\frac{1}{R_e^2} - \frac{1}{(R_e - 0.002)^2} \right)$$

$$\Rightarrow \Delta \vec{F} = \frac{G m_e m}{R_e^2} \left(1 - \frac{1}{0.998^2} \right)$$

$$\Delta \vec{F} = m\left(-0.0393 \tfrac{m}{s^2}\right) = m \Delta a$$

this is a 0.4 % decrease from the standard $\boxed{9.83}$ $\frac{m}{s^2}$ or in other words negligable.

$9.81 \frac{m}{s^2}$?

Next, as long as we are carefull to define our axis (and then measure from that axis) s.t one basis is pointing directly away from the center of the earth, $\vec{F}_g$

What if?

the spring is not always in tension

then there is a possibility that $\vec{F}_s$ becomes < 0 aka not tension any more. could be fine with a stiff rod rather than string

the extention of the spring is due to z & x

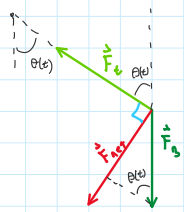
$$\Delta x = \|\vec{r}_2 - \vec{r}_1\|$$

the Pendulums are different lengths

can be a constant vector along said axis.

$$\vec{F}_g = m\langle 0, -9.83\rangle \left(\frac{kgm}{s^2}\right)$$

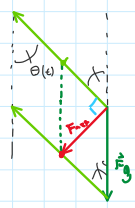
Next is the the tension force, this force is what actually leads to the swinging motion of the pendulum. this swinging gives a hint that may be we should use equations for circular motion.



in order for m to follow a circular path around the center, the net force must be perpendicular to the force of tension.

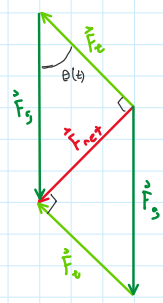
if $\vec{F}_t$ is perpendicular to the acceleration and direction of travel then it can do no work on

we want this tension force to perfectly balance with $\vec{F}_g$, s.t the sum is allways perpindicular to F_t



$$\vec{F}_{net} \cdot \vec{F}_t = 0$$
$$\vec{F}_t + \vec{F}_g = \vec{F}_{net}$$

$\Rightarrow$



$$\frac{o}{h} = \sin(\theta)$$

$$\Rightarrow \frac{\|\vec{F}_{net}\|}{\|\vec{F}_g\|} = \sin(\theta)$$

$$\Rightarrow \|\vec{F}_{net}\| = \|\vec{F}_g\| \sin(\theta(t))$$

$$\Rightarrow \|\vec{F}_{net}\| = mg \sin(\theta(t))$$

$$\Rightarrow \cancel{m}a = \cancel{m}g \sin(\theta(t))$$

$$\Rightarrow \underline{a} = g \sin(\underline{\theta(t)})$$

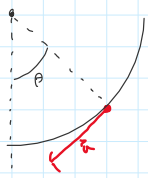
Now we have an expression that describes the magnitude of the net-acceleration of the pendulum. but we have some problems.

1. there are two unknowns: $\theta(t)$, a

2. We lost the direction of acceleration

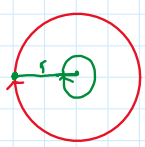
Luckily we can relate $\theta(t)$ & a, and this relationship has the simultanious benifit of re introducing the direction of out vector:

Since we allready know that $\vec{a}$ is causing a circular path we can relate each unit of linear acceleration to a unit of angular acceleration.



Lets start by relating linear distance to angular distance.

Let: θ = angular dist P = Linear dist



$\theta = 2\pi \Rightarrow P = 2\pi r$

the general arc-length relationship can then be diffdemiated to relate changes in position / angle

$$P(t) = \theta(t)\, r$$

$$\Rightarrow \frac{d}{dt}[P] = \frac{d}{dt}[\theta r]$$

assuming $r(\theta) = \text{const}$

$$\Rightarrow P'(t) = \theta'(t)\, r$$

$$\Rightarrow P''(t) = \theta''(t)\, r$$

Since r is a function of neither time nor θ our arc-length relationship directly applies to higher order changes.

$$a(t) = g \sin(\theta(t))$$

$$\Rightarrow \theta''(t)\, r = g \sin(\theta(t))$$

$$a(t) = g\sin(\theta(t))$$

$$\Rightarrow \theta''(t)\, r = g\sin(\theta(t))$$

$$\Rightarrow \ddot{\theta}(t) = \frac{g}{r}\sin(\theta(t)) :$$

$g = -9.81\ \frac{m}{s^2}$

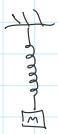
r = Length of Pendulum in m

$\theta(t)$ = angle from vertical @ t

Next, lets find the acceleration due to the spring:

Hook's Law states that an ideal spring produces a force proportional to it's displacement from the equilibrium.

lets imagine a mass hanging from a spring:



the equilibrium of this spring would be the length of the spring that provides enough force to perfectly cancel out F_g

So if you wanted to describe the motion of this mass you could define it relative to the Absolute-equilibrium, or you could describe it relative to the rest-equilibrium

$$\vec{F}_{net} = -k(x(t) - L) + \vec{F}_g$$

$$\vec{F}_{net} = -k\,u(t)$$

- We don't necissarily need to account for all the force produced by the spring, just the imbalance which causes acceleration

So, rather than quantifying the un-forced length of our spring we can just define our motion W.R.T to an initial rest state, making the acceleration due to our spring:

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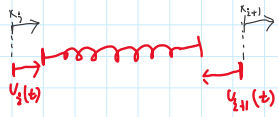


$$\vec{F}_{net} = -k\left(U_i(t) - U_{i+1}(t)\right)$$

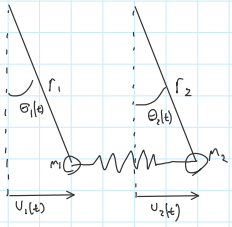
$$\Rightarrow m a_i(t) = k\left(U_{i+1}(t) - U_i(t)\right)$$

$$\Rightarrow a_i(t) = \frac{k_i}{m}\left(U_{i+1}(t) - U_i(t)\right)$$

Very Very bad assumption

Now Lets put it together:

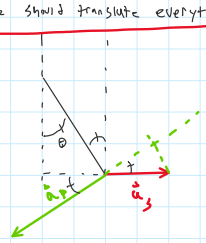
Let's assume that the rest-equilibrium is directly bellow the pivot's of our pendulum:



Now we have a problem, the acceleration from our pendulum is given in terms of an angle and our spring acc is still linear.

as we are mainly working with circular motion, I think we should translate everything into angular coords

Pointless if you just gonna Linearize it anyway.



the component of the spring acceleration that acts on the circular path of the pendulum contributes to the net acceleration

$$\vec{a}_{iT} = a_s \cos(\theta_i(t))$$

the tangent component is only proportional with $\theta_i(t)$ under the assumption that $\vec{F}_s$ is always horizontal

$$\ddot{\theta}_i(t) r = a_s \cos(\theta_i(t))$$

$$\ddot{\theta}_i(t) = \frac{a_s}{r}\cos(\theta_i(t))$$

$$\Rightarrow \ddot{\theta}_i(t) = \frac{k\left(U_{i+1}(t) - U_i(t)\right)}{mr}\cos(\theta(t))$$

the component of the spring force that acts perpendicular to the net force gets canceled out by the

the component of the spring force that acts perpendicular to the net force gets canceled out by the tension force keeping our mass moving in a circular path.

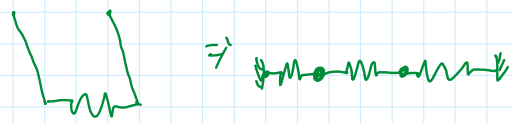
we can now combine the acceleration provided by the pendulum & the acceleration provided by the spring:

$$\ddot{\theta}_i(t) = \ddot{\theta}_p(t) + \ddot{\theta}_s(t)$$

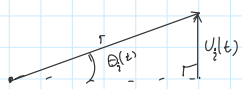
$$\Rightarrow \ddot{\theta}_i(t) = \frac{g}{r}\sin(\theta_i(t)) + K\frac{(U_{i+1}(t) - U_i(t))}{mr}\cos(\theta_i(t))$$

Unfortunately we still have mismatched coordinate systems, thankfully translating from a linear displacement to an angle is something we've already done.

if we're just going to linearize this problem anyway, and the whole point is to start with an approachable version of the coupled mass spring system why not start with the linearization?



$U_i(t) \rightarrow f(t)$; $\theta_i \rightarrow U_i$



$$\sin(\theta) = \frac{o}{h}$$

$$\Rightarrow r_i\sin(\theta) = U_i(t)$$

this final translation gives us our full relationship between angular acceleration and angle w.r.t. t.

$$\ddot{\theta}_i(t) = \frac{g}{r_i}\sin(\theta_i(t)) + \frac{K}{m_i r_i}\left[r_{i+1}\sin(\theta_{i+1}(t) - r_i\sin(\theta_i(t))\right]\cos(\theta_i(t))$$

$\theta_i(t)$ = angular position of ith mass

$\ddot{\theta}_i(t)$ = angular acceleration of ith mass

$g = -9.81$

r_i = length of ith pendulum

K = spring constant

m_i = mass of i

simplifications for 2 mass system:

Let:

$$r_1 = r_2 \;,\; \theta_i(t) << 1 \;\;\&\;\; m_1 = m_2$$

then:

$$\sin(\theta_i(t)) \approx \theta_i(t) \;\;\&\;\; \cos(\theta_i(t)) \approx 1$$

$$\Rightarrow \quad \ddot{\theta}_i(t) = \frac{g}{r}\theta_i(t) + \frac{k}{m}\left(\theta_{i+1}(t) - \theta_{i+1}(t)\right)$$

Why not re-use $\dot{\theta}$

to differentiate the linearized form we'll change θ to X

$$\ddot{X}_i(t) = \frac{g}{r}X_i(t) + \frac{k}{m}\left(X_{i+1}(t) - X_i(t)\right)$$

to conform to convention lets redefine $g := 9.81$

$$\ddot{X}_i(t) = -\frac{g}{r}X_i(t) + \frac{k}{m}\left(X_{i+1}(t) - X_i(t)\right)$$

for: $i \in [1, 2]$

$$\ddot{X}_1(t) = -\frac{g}{r}X_1(t) + \frac{k}{m}\left(X_2(t) - X_1(t)\right)$$
$$\ddot{X}_2(t) = -\frac{g}{r}X_2(t) + \frac{k}{m}\left(X_1(t) - X_2(t)\right)$$

$$\ddot{X}_1(t) = X_1(t)\left(-\frac{g}{r} - \frac{k}{m}\right) + X_2(t)\left(\frac{k}{m}\right)$$

$$\ddot{X}_2(t) = X_1(t)\left(\frac{k}{m}\right) + X_2(t)\left(-\frac{g}{r} - \frac{k}{m}\right)$$

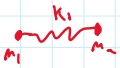
$$\ddot{X}(t) = \begin{bmatrix} \ddot{X}_1(t) \\ \ddot{X}_2(t) \end{bmatrix} \qquad X(t) \begin{bmatrix} X_1(t) \\ X_2(t) \end{bmatrix}$$

$$M = \begin{bmatrix} -\frac{g}{r} - \frac{k}{m}, & \frac{k}{m} \\ \frac{k}{m}, & -\frac{g}{r} - \frac{k}{m} \end{bmatrix}$$

$$\Rightarrow \quad \underset{2\times1}{\vec{\ddot{X}}(t)} = \underset{2\times2}{[M]}\;\underset{2\times1}{\vec{X}(t)}$$

Problems with this diriuation:

1. defining the accelerations for spring separately is wrong if the masses aren't all the same



the force due to a spring at

Define F_s, F_p & F_{net} instead

2. need an in depth transition from ODE to eigenvalue.

3. maybe start with linearization to make vibrational dynamics connection more clear.

how does this effect conservation